quantile_cv.py

Summary

This code defines a QuantileCV class for automated quantile regression tree modeling with Bayesian optimization and pluggable boosting backends.

Dependencies

Standard Library

- typing
- functools

Other

- numpy
- pandas
- pydantic
- sklearn
- xgboost
- catboost

Description

The `quantile_cv.py` file implements a `QuantileCV` class, which provides automated quantile regression using tree-based models and Bayesian optimization. This class is designed for flexible and robust quantile regression tasks, leveraging advanced machine learning techniques.

The implementation includes:

1. `QuantileCV` uses `RegressionCVConfig` for its configuration (same as `RegressionCV`), including:

2. `monotone_constraints`: Dictionary specifying monotonicity direction for variables (0 for none, 1 for increasing, -1 for decreasing)
3. `interactions`: List of lists containing permitted feature interactions
4. `n_jobs`: Number of parallel jobs to use when fitting the model
5. `parameters`: Hyperparameter search space definition (OptimizerParams instance)
6. `return_train_score`: Whether to include training scores during optimization
7. Pre-configured settings (from RegressionCVConfig):
8. `default_regression`: A standard configuration using all hyperparameters
9. `balanced_regression`: A configuration focused on regularization parameters
10. `QuantileCV`: The main class that inherits from `RegressionCV` and provides:
11. Automated hyperparameter tuning via Bayesian optimization
12. Backend selection via `backend` (`"xgboost"` or `"catboost"`)
13. Support for custom quantile levels via the `alpha` parameter
14. Integration with scikit-learn's scoring system for quantile regression metrics
15. Parallel processing support

The class uses cross-validation during the optimization process to ensure model robustness and prevent overfitting. It supports various regression metrics, with a focus on quantile loss, and leverages Pydantic for configuration validation.

Overall, this module offers a comprehensive solution for quantile regression tasks, combining automated model selection, advanced regression techniques, and robust validation mechanisms.